



QUANTITATIVE METHODS 1 · CHAPTER 3

Dispersion and Concentration

Exercise series

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1. Same mean, different risk

Team A daily sales are 48, 49, 50, 51, 52. Team B sales are 30, 40, 50, 60, 70. Compare their means and ranges. What does the comparison show?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

2. Variance and standard deviation

For $x=(2,4,4,6,9)$ calculate the population mean, deviations, squared deviations, variance and standard deviation.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

3. Boxplot and outliers

For an ordered dataset, $Q_1=18$, median =24, $Q_3=30$, minimum =10, maximum =55. Compute the IQR and Tukey fences. Is 55 a potential outlier?

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

4. Comparing relative dispersion

Store A has mean weekly sales EUR 20,000 and SD EUR 3,000. Store B has mean EUR 8,000 and SD EUR 1,600. Compare absolute and relative variability.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

5. Lorenz curve and Gini coefficient

Five households have incomes 10, 10, 20, 20 and 40. Construct cumulative population and income shares, then calculate the Gini coefficient using trapezoids.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

6. Dispersion from a frequency table

The number of weekly complaints per store is distributed as follows:

Complaints x	0	1	2	3	4
Stores n	4	8	10	6	2

- 1 Calculate the mean, population variance and standard deviation.
- 2 Calculate the coefficient of variation.
- 3 Interpret every result in context.

Method prompt. Build columns nx and nx^2 . Use $E(X^2) - [E(X)]^2$ for the population variance.

7. Comparing concentration and robustness

Two markets have the following ordered firm market shares:

Market A	10%	15%	20%	25%	30%
Market B	2%	8%	10%	20%	60%

- 1 Construct cumulative shares for each market.
- 2 Explain which Lorenz curve is farther from equality.
- 3 For each market calculate the range and discuss why it is insufficient.
- 4 State a defensible concentration conclusion.

Method prompt. With five equal-weight firms, cumulative population shares increase by 20 percentage points.